Kvist og Bjælke

Network design

ernvervsakaaemi

2.1 The Company's network

Kvist & Bjælke ApS has several employees and IT systems that must operate within the company's network.

The company currently has the following devices:

Device	Quantity	Description
Desktop computers	15	Workstations used for administration and office tasks
Laptops	10	Used by employees working from different locations
Network printers	3	Shared printers available to employees
Servers	3	Hosting internal systems and applications
IP phones	12	Telephony using the company's network
Wireless access points	4	Wireless network access for employees and guests
Surveillance cameras	6	IP-based video surveillance
Network equipment	As required	Switches, routers, and other necessary network equipment
Guest devices	Variable	Smartphones and computers belonging to visitors

ID	Name	Devices	Subnet	Gateway	DHCP
70	Office VLAN 1	Desktop Computers: 15 Laptops : 10 Total devices: 25	10.5.70.2/26	10.5.70.2	10.5.70.2 2 to .70.62
73	Office VLAN 3	Network printers servers :3 Total devices: 3	10.5.73.2/29	10.5.73.2	10.5.73.2 2 to .73.62

72	Office VLAN 2	Surveillance cameras Network equipment: 6 Total devices: 6	10.5.72.2/28	10.5.72.2	None(Static)
71	Office VLAN 3	Servers: 3	10.5.71.2/29	10.5.71.2	None(Static)
20	Phone VLAN 1	Ip Phones: 12	10.5.20.2/27	10.5.20.2	10.5.20.2 2 to .20.32
30	Lobby VLAN 1	Guest devices	10.5.30.2/26	10.5.30.2	10.5.30.2 2 to .30.62
31	Lobby VLAN 2	Employees	10.5.31.2/29	10.5.31.2	10.5.31.2 2 to .31.6

Subnet decisions

The reason we have chosen the specific subnet masks, is that there could be hired some new employees soon. The other reason is that we expect the company to expand in the next few years, so that the company will have a buffer for the future. That would also make so we don't have to touch the network configuration in the near future.

Network devices

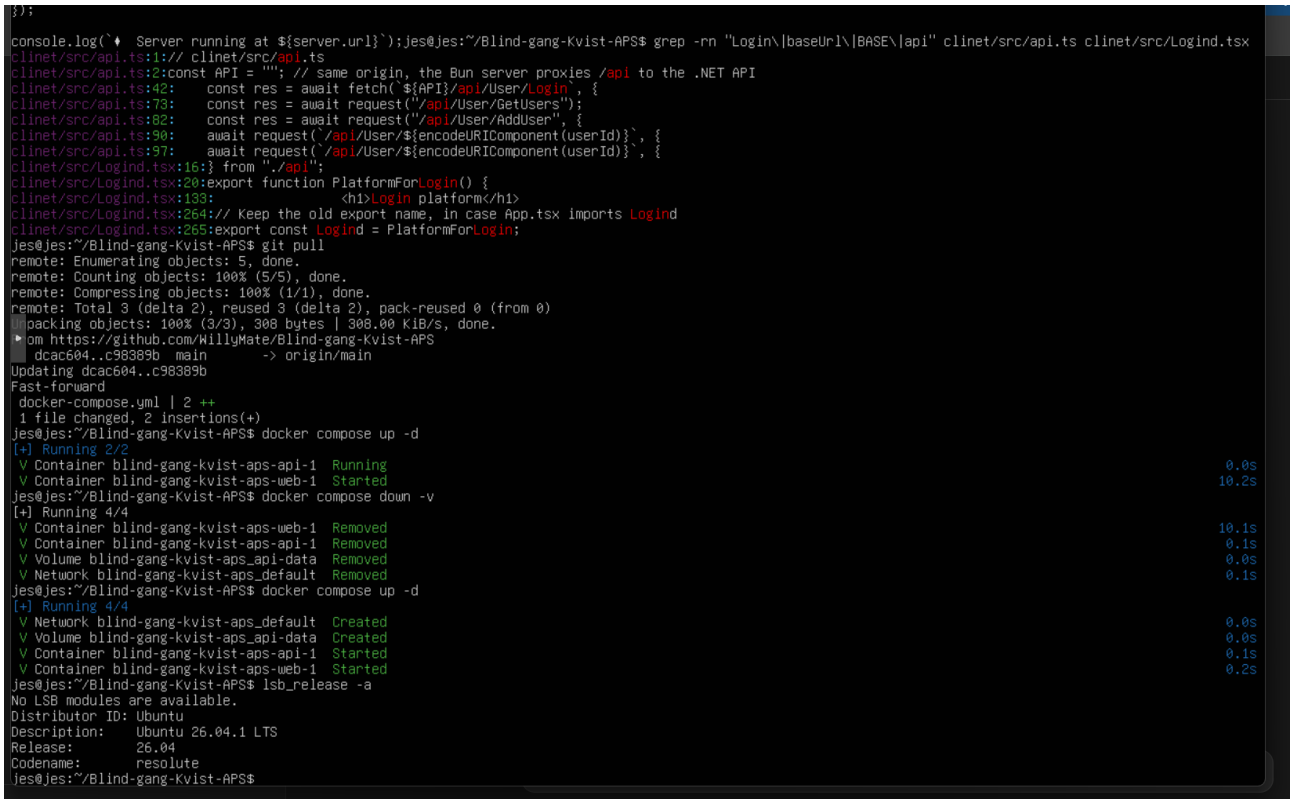
We have decided that the company needs one firewall for the network. The firewall is setup so it only allows data that follows its configuration. The router is a part of the firewall so it's a firewall/router combo.

We will have four access points in total. We will be having two in the office, one in the lobby and one meetings room.

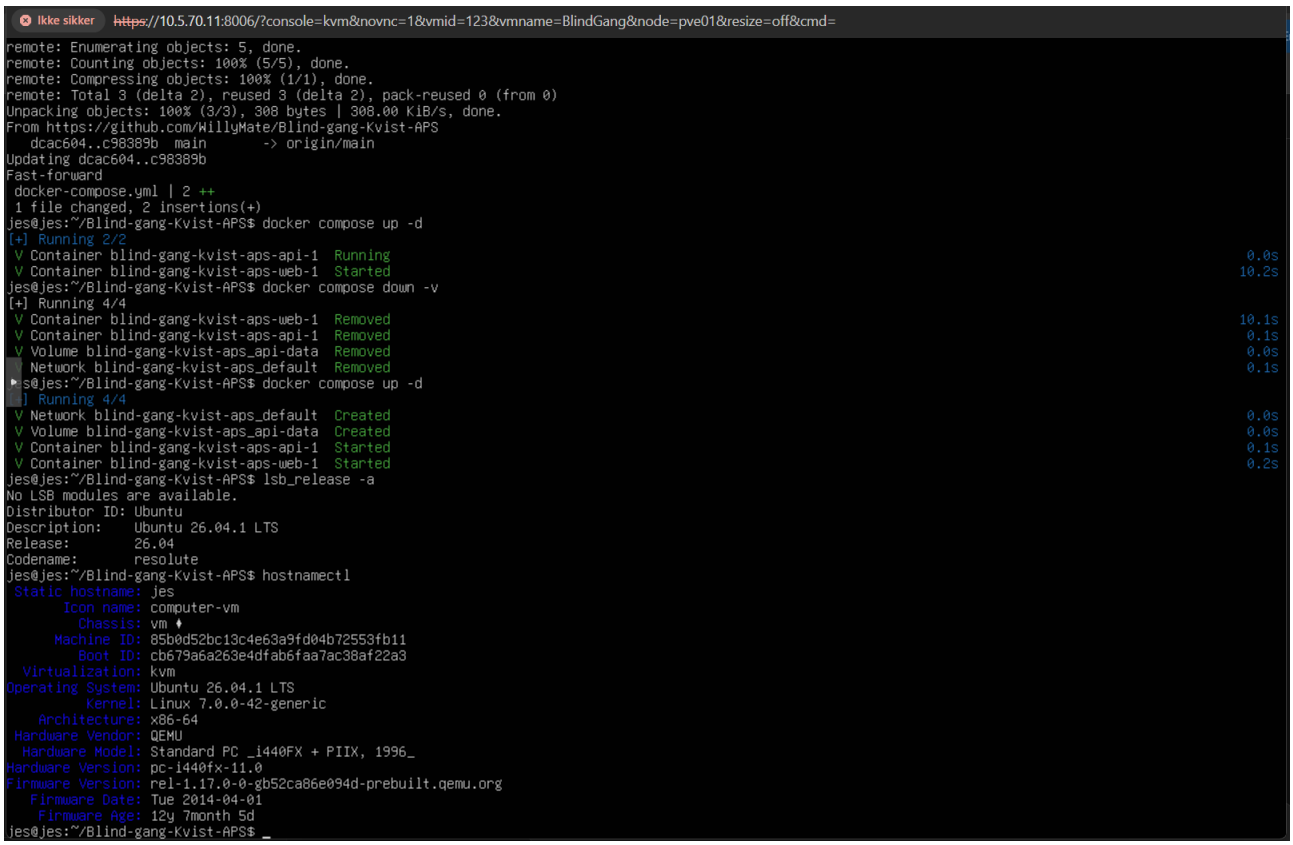
The amount of switches we want is around 48 port unit cause a 24 port unit would be too close to the bare minimum.

Testing and Documentation

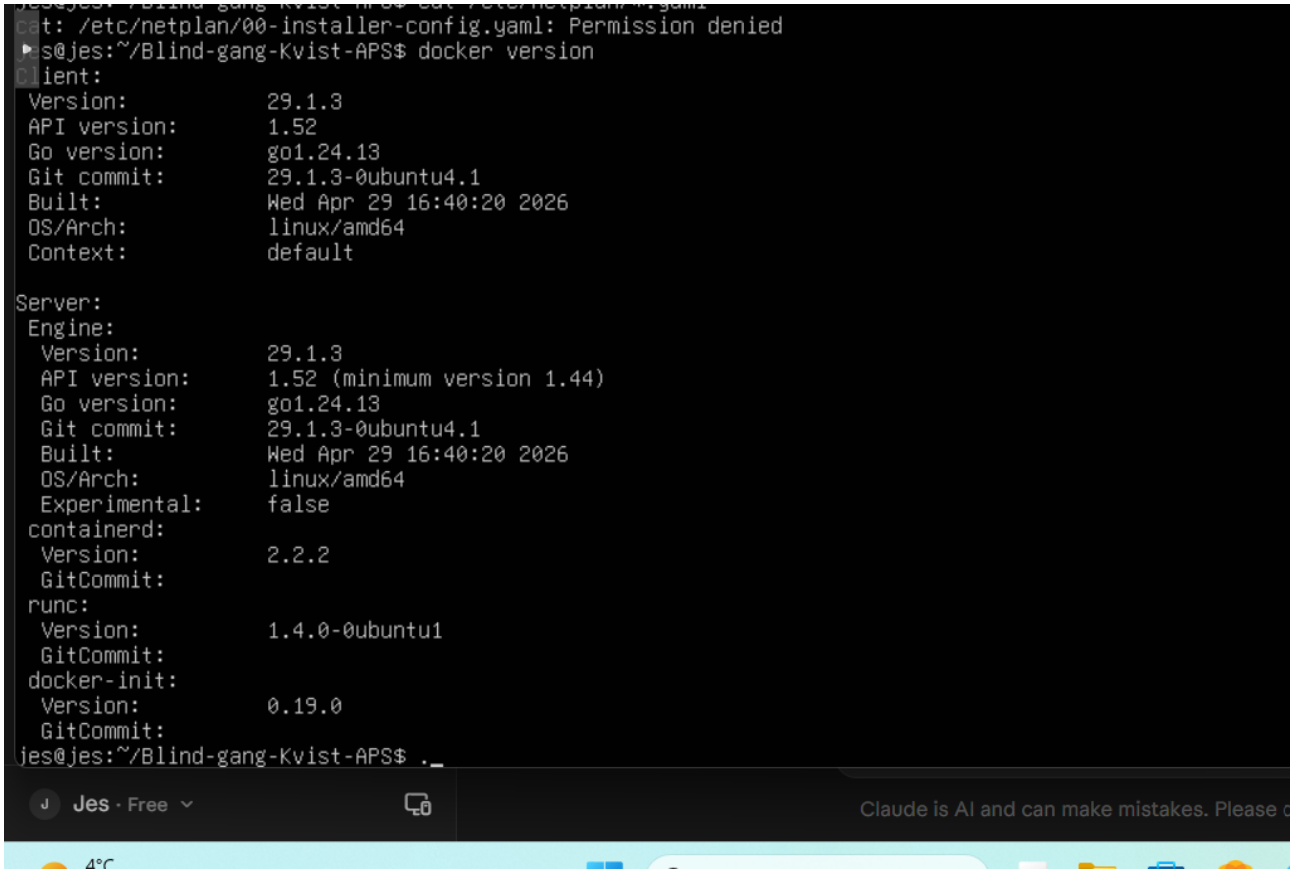
Server – VLAN 571



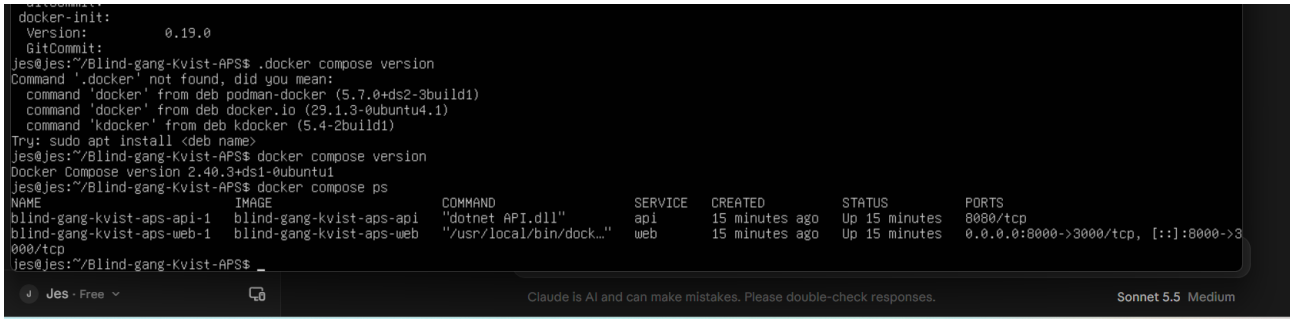
This image show that the Ubuntu server is installed.



This image shows part of the Ip configuration.

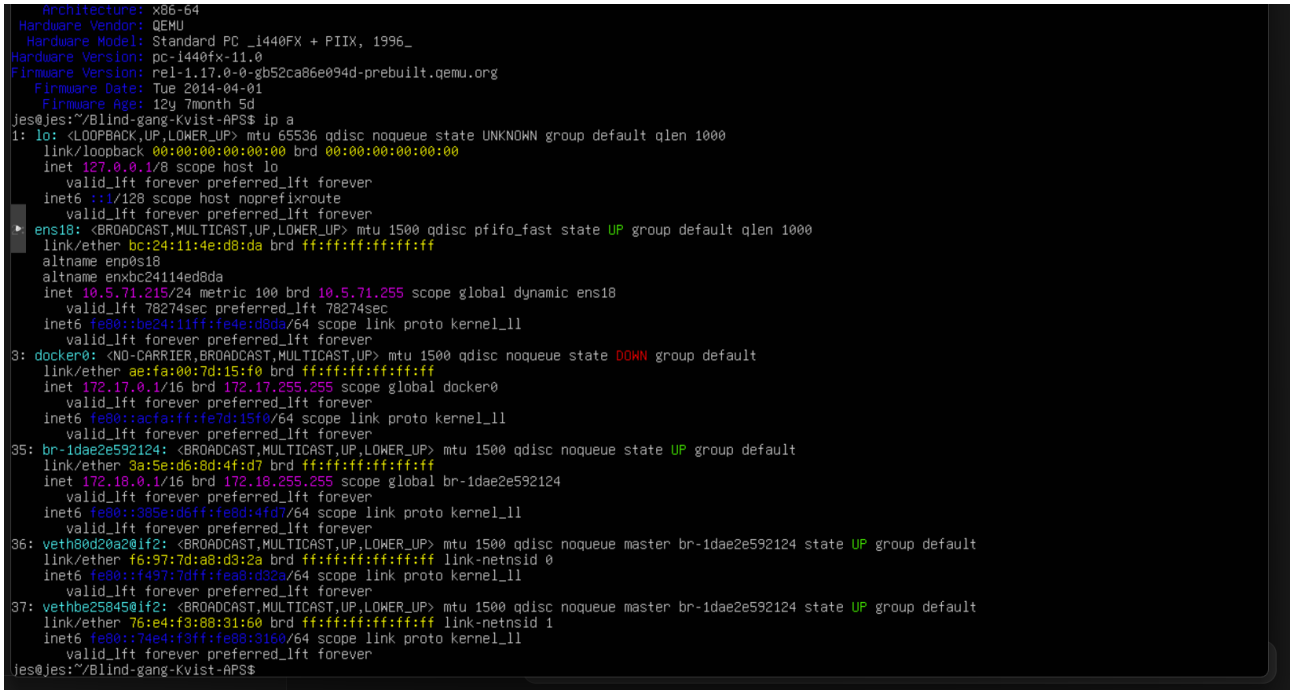


This image shows That the docker have been installed and what version it is using.

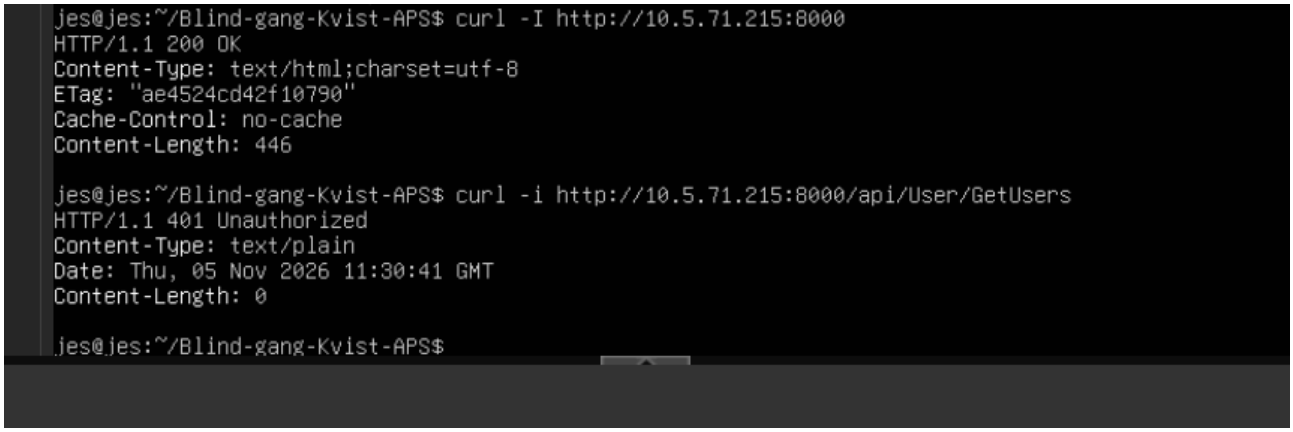


This image shows the login platform running.

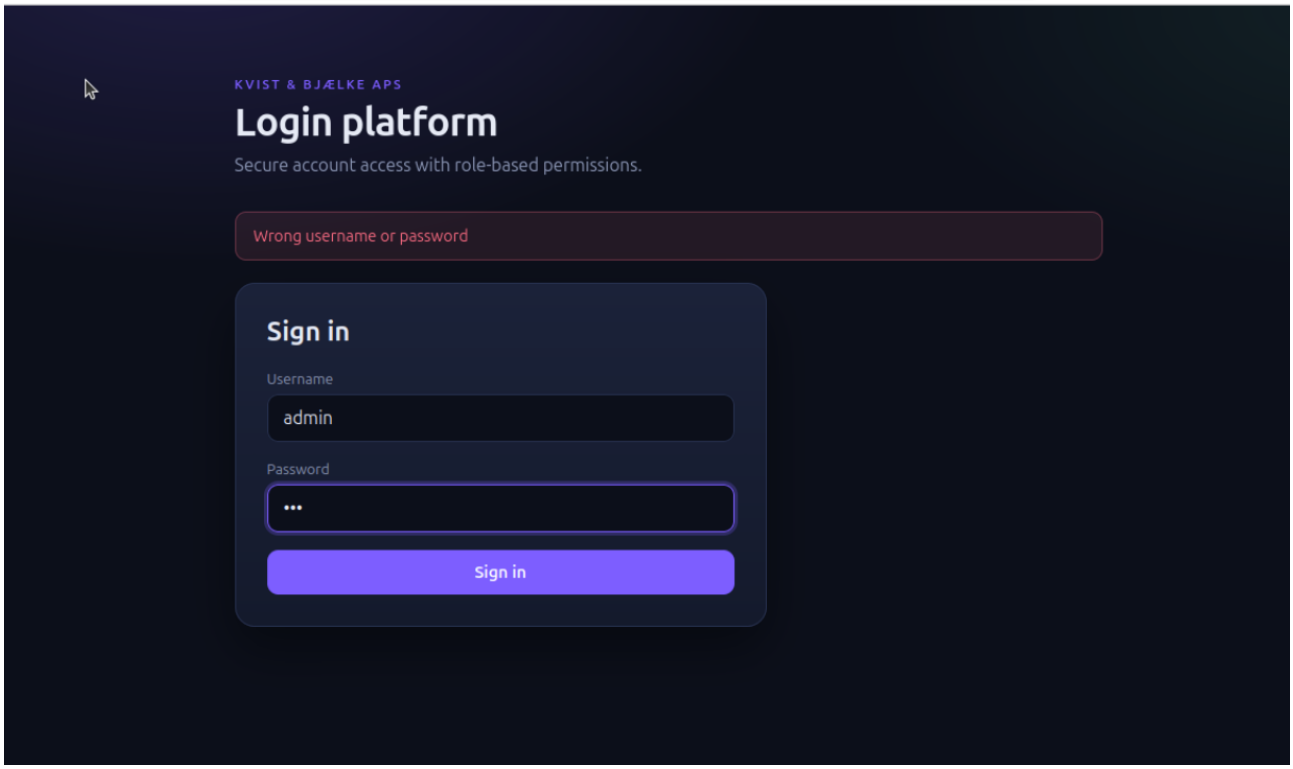
Client – VLAN 570



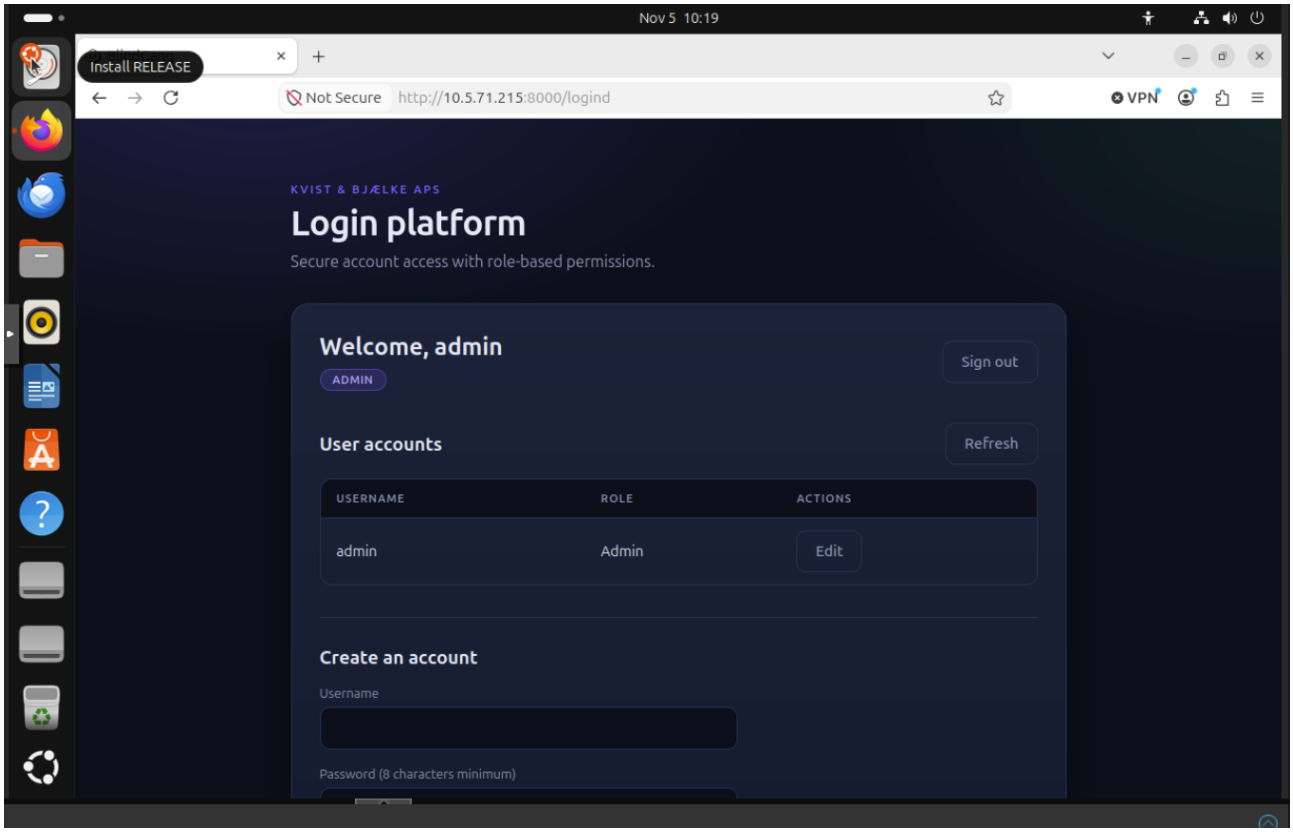
This image shows that the client has the correct configurations.



This image shows that we have a security features.



This image shows when you give wrong login details.



Shows how it looks when you locked in.

On it shows that the docker container is running since we are using it on a browser.

Short description of the general program.

1.

Frontend: Is a react client served by a bun server we have sat up. That is our login page and our user management view.

Backend: Our backend is our server part where we have used ASP.NET Core. The web API is split into three projects. The API that controls authentication setup. Service our business logic such as the passwordhasher.

Deployment: Both of the parts has their own docker container and started with docker compose.

The web container is published on host port 8000, and the API is only reachable inside dockers internal network.

2. Some of the security features we have used is passwordhashing and cookie authentication. Passwordhashing works so you don’t store the password in the table and can`t be decrypted. Cookie authentications work so when a user logs in that it creates an encrypted authentication cookie. The browser stores the cookie and sends it automatically with every later request. The server will decrypt the cookie. When the user logout the cookie clears.

3. GitHub repository link : https://github.com/WillyMate/Blind-gang-Kvist-APS

Struggles and problems

We had some problems with the tokens, because it something new we haven’t tried to use that much yet. So we went back to some of the materials that our teacher has given us about that topic, we also used AI with some of the stuff we didn’t understand. We also asked some of our fellow students about how they had done it. So we had many different sources of information to help us.

Final reflection

It was a great new experience of how to get these to classes to work together, and find out how much they actually lean up against each other. We found out that there is many layers to securing a safe project, and you can make it more safe with the internet infrastructure. It’s the same you can say about networking, we ran into a lot of problems with deploying to Proxmox because we had setup our Dockerfiles wrong, so we had to go back into the code. If it was a real company we would be a lot more careful about who we share our stuff with, and how we approach the networking.